FINAL PROTOTYPE PRESENTATION

CityServe

Premium local services, made convenient for Thapar

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Local services are hard to find and hard to trust

Fragmented

Electricians, tutors and mechanics are mostly found through word of mouth.

Hard to verify

Listings and ratings are rarely labelled by how trustworthy they are.

Disconnected

Search, calling, navigation and booking live in separate places.

The prototype at a glance

Real, sourced Patiala data combined with a working ordering, review and admin system.

155

Patiala businesses sourced via the Google Places API

20

Trade categories, from electricians to pest control

83

Patiala localities for filtering

3

User roles: customer, vendor, admin

Three-tier architecture

Frontend

Next.js 14 App Router, TypeScript and Tailwind. Server-rendered homepage, Mapbox map with fallback. Port 3000.

Backend API

Express and TypeScript with Sequelize and mysql2. JWT auth, Zod validation, Helmet, rate limiting. Port 4000.

Data layer

MySQL 8 for users, orders and reviews, plus the 155-record Patiala dataset held in memory. Port 3306.

Data model: six core tables

User

Customer, vendor or admin account.

Category

Trade grouping for vendors and offerings.

Vendor

Business profile with location, phone and verification flag.

Offering

A product or a service sold by a vendor.

Order

Pending, approved, rejected, completed or cancelled, with payment status.

Review

One review per order, tied to vendor and offering.

Features by user role

Customer

Browse, search and filter. Order products or book services, track status, then review a completed order.

Vendor

Register a business, manage products and services, and approve, reject or complete incoming orders.

Admin

See all users and vendors, toggle vendor verification and view platform analytics.

Change summary since the initial prototype

Area	Initial prototype	Final prototype
Data	Dataset file unused	155 real Patiala businesses live in the app
Backend	Mock vendors, fragile queries	Database and dataset merged safely
Look and feel	Generic template, placeholder cards	Heritage-inspired editorial design
Ratings	Default 5.0 stars	Honest "Not rated yet"
Actions	Browse only	One-tap call, directions, map view

What users see now

Find faster

New hero search with query, category and 83-locality filters. Live debounced results on Explore, with removable filter chips.

Contact instantly

Every vendor profile has a one-tap call button and Google Maps directions. Explore also has a map view toggle.

Trust at a glance

Verified Partner and Directory Listing badges. Missing ratings read "Not rated yet", never a fake 5.0.

In-memory dataset engine

RFC 4180 tokenizer

Hand-written state machine. Handles quoted fields, commas inside addresses and \r\n line endings.

Singleton cache

Parsed once at server boot and reused. No file reads on repeat calls (under 2 ms per project report).

Typed DTO

Each row becomes a BackendVendorDto with nullable phone, rating and review count, plus locality and category indexes.

Constraint: zero schema changes or migrations. Code: dataset.service.ts and frontend/src/lib/dataset.ts.

Hybrid backend controllers

vendor.controller

listVendors queries MySQL via Sequelize and the dataset, merges both, verified partners first.

getVendorById

Primary-key lookup, then fallback to place_id, URL-encoded slug or normalized name.

category.controller

Merges database and dataset categories with live counts, e.g. Laundry 15.

search.controller

Op.like on offerings and vendors plus in-memory fuzzy match on name, category, subcategory and area.

Design system: The Local Standard

Deep Forest
#183628

Warm Ivory
#FAF9F6

Soft Sage
#E8EEE7

Muted Brass
#B79A64

Charcoal Ink
#202321

DM Serif Display for headlines and Inter for interface text. Replaces the generic blue SaaS look, with light and dark themes.

Dataset coverage

Category group	Listings
Laundry / Dry Cleaning	15
Car/Bike Mechanics	10
11 trades, e.g. Electricians, Plumbers, Tutors	8 each
4 trades, e.g. Painters, RO Service	7 each
Appliance Repair, Pest Control, Cleaning	5, 5, 4

4.82

Mean listing rating across 148 rated businesses

22,119

Google reviews behind those ratings

Integrity and bug fixes

Honest data

RatingStars handles null without NaN. 7 records have no rating and 14 have no phone, and the UI says so plainly.

Theme contrast fix

```
:root {
color-scheme: light;
}
.dark {
color-scheme: dark;
}
```

Root cause

Native inputs followed the OS theme, not the app's .dark class, giving dark text on dark backgrounds.

Build and API verification

Check	Result
next build	Compiled, types valid, 24 of 24 pages generated
Shared first-load JS	87.3 kB
GET /api/health	status ok, payments and maps in demo mode
GET /api/vendors	155 combined listings
GET /api/categories	20 categories with counts
GET /api/search?q=electric	10+ matching Patiala vendors

Known gaps and next steps

Technical debt

Parser duplicated in frontend and backend. Explore ships 611 kB first-load JS. README describes more tables than the six models in code.

Next steps

Wire the recommendation score (proximity, rating, trust) into ranking. Add importer and API tests. Claim-this-listing flow. Product-type dataset support.